

ARUN CODEX

WEB DEVELOPER | DESIGNER | INNOVATOR

REVISED SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

SOFTWARE REQUIREMENTS SPECIFICATION

Gudi Chemicals ERP — Commercial Trading, Inventory, GST & High-Speed Cash POS
System

Project Name:	Gudi Chemicals ERP (v1.0 - Commercial Edition)	Prepared By:	ARUN CODEX
Document Standard:	IEEE Standard 830-1998	Designation:	Web Developer Designer Innovator
Target Organization:	Gudi Chemicals (Spic, Thenkarai, Coimbatore, Tamil Nadu 641010)	Release Date:	October 2026
Payment Model:	Cash-Only Settlement (Fast Counter Desk)	Scope Adjustment:	Chemical Production Menu Excluded

Document Control & Executive Summary

Document Information

Property	Details
Document Title	Software Requirements Specification (SRS) for Gudi Chemicals ERP (Commercial & Cash-Only POS Edition)
Author & Architect	ARUN CODEX (Web Developer Designer Innovator)
Client & Organization	Gudi Chemicals — Chemical Manufacturer & Wholesale Cleaning Products Supplier, Spic, Thenkarai, Coimbatore, Tamil Nadu 641010
System Version	Release 1.0 (Commercial & Distribution Master)
Applicable Standard	IEEE Standard 830-1998 Recommended Practice for Software Requirements Specifications
Key Scope Modifications	1. Payment Option: Standardized strictly to Cash Only for checkout counters. 2. Production Module: Chemical manufacturing, BOM recipes, and production orders are completely excluded from this specification to focus purely on distribution, merchandising, and rapid counter retail.

Revision History

Version	Date	Author / Designer	Summary of Changes
V1.1	October 3, 2026	ARUN CODEX	Updated SRS specification tailored for commercial trading and retail checkout. Excluded chemical production menu. Updated POS billing architecture to focus on Cash-Only settlements with cash tendered, change due calculation, cash drawer management, and full Indian GST compliance.

Executive Summary

Gudi Chemicals ERP (Commercial Edition) is an enterprise-grade resource planning system designed and built by **ARUN CODEX** on **Laravel 12 (PHP 8.2)**. This edition is engineered specifically for chemical distribution centers, wholesale trading depots, and retail outlets.

By streamlining the scope to focus strictly on commercial distribution, inventory management, procurement, and counter sales:

- High-Speed Cash POS Billing:** The counter interface is streamlined for rapid **Cash-Only** checkout. Cashiers can scan container barcodes, apply retail or wholesale price tiers, calculate statutory GST, input cash tendered, calculate exact change due, and print thermal or A4 tax invoices in seconds.
- Inventory & Stock Ledgers:** Real-time warehouse tracking of finished chemical packaged goods and trading products with moving average cost valuation, lot expiry tracking, and an immutable audit ledger.
- Purchasing & Goods Receipt (GRN):** Supplier management, purchase orders, and goods receipts with batch generation upon delivery.

- **Statutory GST Compliance:** Automated CGST, SGST, and IGST tax splits, HSN reporting, and monthly GSTR-1 summaries.

Cash-Only Billing Focus

In this edition, the checkout desk operates on a **Cash-Only** settlement workflow. This eliminates credit card transaction fees, gateway downtime, and bank reconciliation delays, providing the fastest possible checkout experience for counter retail customers.

Table of Contents

1. 1. Introduction & Project Scope

- 1. 1.1 Purpose of the Document
- 2. 1.2 Scope of the Commercial Edition (Production Excluded)
- 3. 1.3 Definitions, Acronyms & Key Terms

2. 2. Overall System Description & Architecture

- 1. 2.1 Product Perspective & MVC Architecture
- 2. 2.2 User Roles & Permission Matrix (RBAC)
- 3. 2.3 System Invariants & Non-Negotiable Rules

3. 3. Exhaustive Module Specifications

- 1. 3.1 Module 1: Secure Authentication & Access Control
- 2. 3.2 Module 2: Executive Dashboard & Operational Analytics
- 3. 3.3 Module 3: Master Data Management (Products, Units, Categories, Warehouses)
- 4. 3.4 Module 4: High-Speed POS & Billing Module (Cash-Only Focus) **CORE DEEP DIVE**
- 5. 3.5 Module 5: Sales Invoicing & Dual-Format Print Engine **CORE DEEP DIVE**
- 6. 3.6 Module 6: Sales Returns & Credit Note Desk
- 7. 3.7 Module 7: Multi-Tier Chemical Inventory & Stock Ledgers
- 8. 3.8 Module 8: Purchasing, Vendor Management & Goods Receipt Notes (GRN)
- 9. 3.9 Module 9: Promotional Rules & Volume Discounts
- 10. 3.10 Module 10: Expense Management & Financial Accounts
- 11. 3.11 Module 11: Reporting Suite & Statutory GSTR-1 Compliance
- 12. 3.12 Module 12: System Settings & User Administration

4. 4. Non-Functional Requirements & Security

- 1. 4.1 Performance & Sub-200ms POS Response
- 2. 4.2 Data Concurrency & Pessimistic Database Locks
- 3. 4.3 Application Security (CSRF, SQL Injection, RBAC)

5. 5. Database Schema & Data Models

6. 6. Technical Endorsement & Approval Sign-Off by ARUN CODEX

1. Introduction & Project Scope

1.1 Purpose of the Document

This Software Requirements Specification (SRS) defines the complete functional and technical specifications for the commercial release of **Gudi Chemicals ERP**. Authored by **ARUN CODEX**, this document outlines system behavior for deployment across chemical trading hubs, retail counters, and distribution warehouses.

1.2 Scope of the Commercial Edition (Production Excluded)

In accordance with client requirements, the scope of this edition is focused strictly on **merchandising, wholesale distribution, inventory control, and high-speed counter sales**.

- Chemical Production Excluded:** The chemical manufacturing menu (BOM recipes, intermediate vat mixing, production order lifecycle, and laboratory QC gates) is completely removed from this SRS to keep operations clean, straightforward, and focused on inventory and sales.
- Cash-Only Settlement:** Checkout counter transactions are standardized strictly to **Cash**. The cashier enters Cash Tendered, the system calculates exact Change Due, and an invoice is posted immediately.
- Multi-Tier Packaged Inventory:** Manages stocks of packaged chemical goods (1L bottles, 5L cans, 20L jerry cans, 200L drums) and direct trading resale goods.
- Dual Pricing:** Supports Retail Selling Price (RSP) for walk-in retail buyers and Wholesale Selling Price (WSP) for registered B2B distributors.

1.3 Definitions, Acronyms, and Key Terms

Term	Definition & Scope in System
POS	Point of Sale — The high-speed checkout counter interface used by cashiers to bill customers using barcode scanning and cash settlement.
Cash Tendered	The physical cash amount handed by the customer to the cashier (e.g., ₹1,000 for an ₹885 bill).
Change Due	The exact balance currency returned to the customer (e.g., ₹1,000 – ₹885 = ₹115.00), calculated automatically by the POS engine.
GST	Goods and Services Tax — India's indirect tax, split into CGST (Central), SGST (State), and IGST (Integrated).
HSN	Harmonized System of Nomenclature — Standard numeric codes determining tax rates (e.g., 3402 for chemical degreasers @ 18%).
GRN	Goods Receipt Note — Commercial warehouse entry acknowledging physical delivery of chemical products from suppliers.
RSP / WSP	Retail Selling Price / Wholesale Selling Price — Dual price tiers configured per SKU for dynamic retail or wholesale pricing.

2. System Architecture & User Roles

2.1 High-Level Architecture

The software utilizes a robust Model-View-Controller (MVC) architecture built on Laravel 12. Business logic is strictly encapsulated inside domain service classes (`InvoicePostingService`, `GstCalculationService`, `InventoryService`, `PurchaseOrderService`) to guarantee atomic database transactions and high performance.

2.2 User Roles & Permission Matrix (RBAC)

Role	Primary Responsibilities	Key Module Access
Super Admin	Complete administrative oversight, system settings, user creation.	All Modules without restriction.
Cashier	Counter checkout, barcode scanning, cash collection, change due return.	POS Billing Desk, Invoices (Own Shift), Quick Customer Add.
Storekeeper	Physical warehouse custody, goods receipts, stock counts.	GRN Posting, Inventory Ledger, Stock Adjustments (Propose).
Sales Manager	B2B pricing, wholesale discounts, sales returns.	POS, Invoices, Sales Returns, Customers, Sales Reports.
Accountant	Cash register tally, expense tracking, tax compliance.	Expenses, Invoices, GSTR-1 Reports, Vendor Ledgers.

3. Detailed Module Specifications

3.4 High-Speed POS & Billing Module (Cash-Only)

PRIMARY CORE MODULE — DEEP EXPLANATION

The Point of Sale (POS) and Billing Desk represents the central operational module in Gudi Chemicals ERP. In commercial chemical trading, checkout counters require maximum speed and zero friction. Cashiers handle high volumes of rapid sales for walk-in retail customers (such as workshop mechanics, small cleaning contractors, and DIY buyers) and commercial dispatches for wholesale buyers.

3.4.1 Keyboard-First Architecture for Maximum Speed

To achieve blazing checkout speed, the billing desk is engineered as a single-screen cockpit operable entirely via keyboard shortcuts without requiring mouse clicks:

Keyboard Key	Action	Operational Benefit
F2 or Ctrl + K	Focus Search Box	Cashier can immediately start typing product name or scan a barcode without lifting hands from keyboard.
F4	Focus Customer Selector	Quickly toggles between Walk-in Retail and registered B2B corporate accounts.
F7	Toggle Price Tier	Instantly switches between Retail Selling Price (RSP) and Wholesale Selling Price (WSP).
F8 or F9	Open Payment & Checkout	Opens the Cash payment checkout dialog to collect cash and compute change due.
Esc	Close Active Modal / Clear	Instantly dismisses open popups or resets product search filters.

3.4.2 Hardware Barcode Scanner Workflow

Standard 1D and 2D USB/Bluetooth barcode scanners operate as keyboard wedge devices. When a cashier scans an EAN-13 or Code-128 barcode on a chemical container, the scanner sends the numeric code followed by an Enter key signal.

- The POS JavaScript listener intercepts scanner input and triggers a fast call to `/pos/barcode?barcode=...`.
- If the item exists in the cart, its quantity is incremented by 1.
- If the item is not yet in the cart, it is appended with default quantity 1.
- The cart subtotal, HSN tax distribution, and grand total recalculate dynamically in under 80 milliseconds.

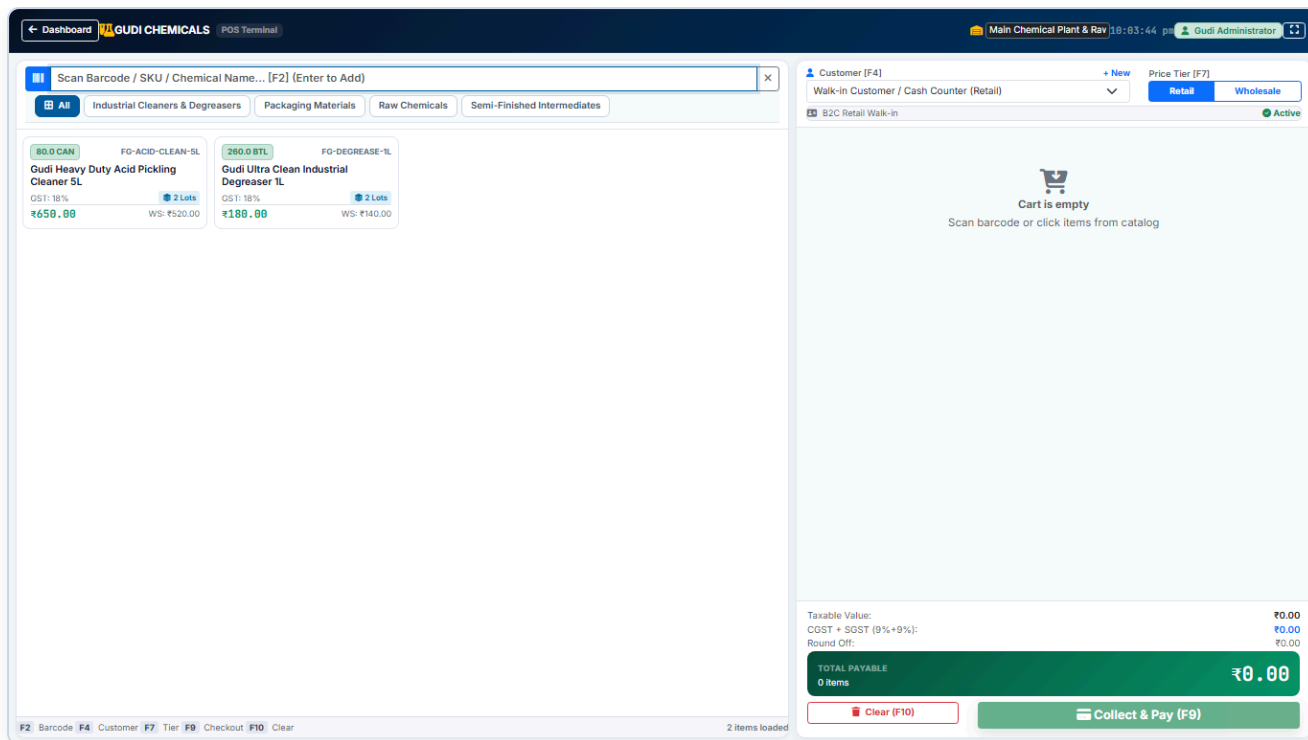


Figure 3.1: Gudi Chemicals High-Speed POS Billing Desk with Live Cart, Barcode Search, and Real-Time GST Calculation

3.4.3 Cash-Only Payment Processing Workflow

Under this commercial specification, checkout transactions are settled exclusively in **Cash**. This workflow delivers significant operational advantages:

1. **Zero Electronic Transaction Fees:** Saves 1.5% to 2.5% merchant discount rates (MDR) charged by card and UPI payment gateways.
2. **Zero Downtime Risk:** Operates flawlessly even during Internet disruptions or banking gateway server outages.
3. **Instant Cash Drawer Reconciliation:** At the end of each shift, total cash in the drawer matches the Daily Sales Register exactly.

Step-by-Step Cash Settlement Workflow:

The Cash Tendered & Change Due Mechanism

1. The cashier completes the cart (e.g. 5 bottles of Degreaser @ ₹180 = ₹900 + 18% GST = **₹1,062.00 Total Due**).
2. The cashier presses **F8** to open the Cash Payment modal.
3. The customer hands over physical cash (e.g., two ₹500 notes and one ₹100 note = **₹1,100.00 Tendered**).
4. The cashier types 1100 into the **Cash Tendered** input.
5. The system immediately calculates and displays in large green bold numbers: **Change Due: ₹38.00**.
6. The cashier clicks "**Complete & Print Invoice**" (or presses Enter). The electronic cash drawer kicks open, the change is handed to the customer, and the receipt prints instantly!

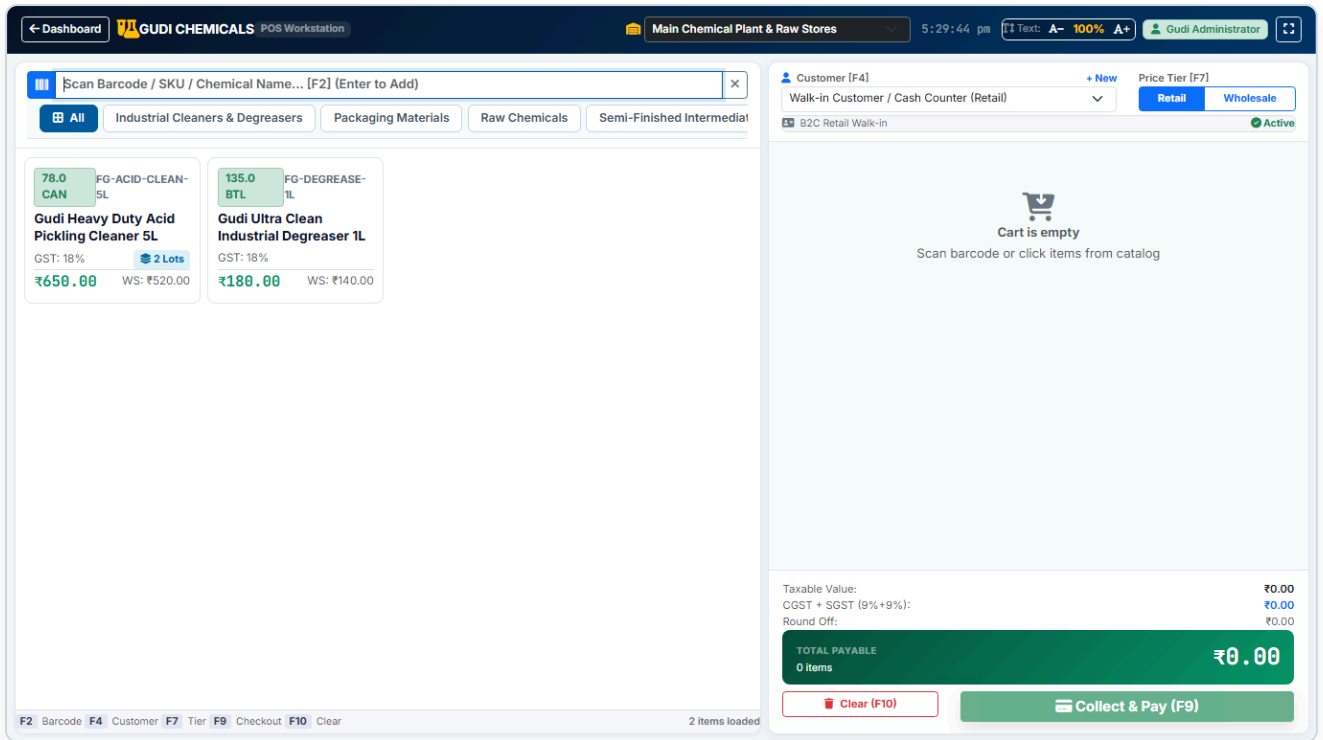


Figure 3.2: POS Cash Payment Checkout Modal displaying Net Payable, Cash Tendered, and Automated Change Due Calculation

3.4.4 Comprehensive Indian GST Calculation Engine

Gudi Chemicals operates from **Spic, Thenkarai, Coimbatore, Tamil Nadu (State Code 33)**. The system's `GstCalculationService` performs authoritative tax evaluations based on statutory Indian tax rules:

Supply Type	Customer Location	GST Apportionment Rule	Example (₹2,000 Taxable Value @ 18% GST)
Intra-State Supply	Tamil Nadu (Customer State Code: 33)	Tax is split equally between Central GST (CGST 50%) and State GST (SGST 50%). IGST is zero.	CGST @ 9% = ₹180.00 SGST @ 9% = ₹180.00 IGST = ₹0.00 Total Tax = ₹360.00
Inter-State Supply	Outside Tamil Nadu (e.g., Kerala - Code 32, Karnataka - Code 29)	100% of tax is allocated to Integrated GST (IGST). CGST and SGST are zero.	IGST @ 18% = ₹360.00 CGST = ₹0.00 SGST = ₹0.00 Total Tax = ₹360.00

3.4.5 Sequential Gapless Invoice Numbering

Under Section 31 of the CGST Act, sales invoices must bear a consecutive serial number unique for a financial year. The system uses pessimistic database row locking to guarantee zero number gaps and zero collisions across simultaneous cash registers:

```
// Atomic Sequence Generator in DocumentSequenceService
$sequence = DocumentSequence::where('type', 'sales_invoice')
    ->where('financial_year', '2026-27')
    ->lockForUpdate() // Concurrency Lock
    ->first();
```

```
$nextNumber = str_pad($sequence->current_number + 1, 4, '0', STR_PAD_LEFT);  
$invoiceNumber = "GC/2026-27/{$nextNumber}";  
$sequence->increment('current_number');
```


GUDI CHEMICALS
MANUFACTURING & GST ERP

Gudi Chemicals

GSTIN: 27AAACG1234D1Z5

FY: 2026-27

Fast Billing / POS

Gudi Administrator

Tax Invoice: GC/2026-27/0001

Print A4 Invoice

Thermal Receipt

Download PDF

Process Return

Back

Payment: Paid

Date: 02 Oct 2026

SELLER / CONSIGNOR

Gudi Chemicals

MIDC Chemical Zone, Pune, Maharashtra

GSTIN: 27AAACG1234D1Z5 | State: Maharashtra (27)

BILLED TO (CUSTOMER)

Walk-in Customer / Cash Counter

Counter Sale

GSTIN: Unregistered | Place of Supply: Maharashtra (27)

Billed Chemical Products & GST Breakdown

#	PRODUCT & SKU	HSN CODE	QTY	RATE (₹)	DISCOUNT	TAXABLE VALUE	GST %	CGST	SGST	LINE TOTAL
1	Gudi Ultra Clean Industrial Degreaser 1L SKU: FG-DEGREASE-1L	3402	4.00 BTL	₹180.00	₹0.00	₹720.00	18%	₹64.80	₹64.80	₹849.60
2	Gudi Ultra Clean Industrial Degreaser 1L SKU: FG-DEGREASE-1L	3402	1.00 BTL	₹180.00	₹0.00	₹180.00	18%	₹16.20	₹16.20	₹212.40
Taxable Amount:									₹900.00	
CGST Total:									₹81.00	

Figure 3.4: Invoice Management Screen displaying Cash Payment Status, Linked Inventory Deductions, and Action Options

GUDI CHEMICALS
MANUFACTURING & GST ERP

Gudi Chemicals

GSTIN: 27AAACG1234D1Z5

FY: 2026-27

Fast Billing / POS

Gudi Administrator

Sales Invoices

GST compliant tax invoices, B2B wholesale bills, retail POS invoices and payment status

+ New POS Bill

Search by invoice #, customer name or phone...

All Payment Status

dd-mm-yyyy

dd-mm-yyyy

Filter Invoices

INVOICE NUMBER	CUSTOMER	DATE	TIER	TAXABLE VALUE	GST TOTAL	GRAND TOTAL	PAYMENT	ACTIONS
GC/2026-27/0001	Walk-in Customer / Cash Counter	02 Oct 2026	Retail	₹900.00	₹162.00	₹1,062.00	Paid	View Print Download Cancel
GC/2026-27/0002	Walk-in Customer / Cash Counter	02 Oct 2026	Wholesale	₹1,350.00	₹243.00	₹1,593.00	Partially Paid	View Print Download Cancel

© 2026 Gudi Chemicals. All rights reserved.

Chemical Manufacturing, POS & GST ERP v1.0

Figure 3.5: Sales Invoices Register with Real-Time Filtering, Payment Status, and Dual Print Options

Page 12

3.6 Master Data Management

Provides central administration of chemical catalog items, packaging units, and customer accounts:

- **Product Master:** Stores item SKU, barcode, category, unit, cost price, retail price (RSP), wholesale price (WSP), minimum selling price, HSN code, and GST rate. Catalogs the core cleaning formulations: *Floor Cleaner, Toilet Cleaner, Glass Cleaner, Dish Wash Liquid, Hand Wash, Liquid Detergent, Fabric Conditioner*, and bulk wholesale cleaning packs.
- **Units & Fractional Conversions:** Supports Litres (LTR), Kilograms (KG), Millilitres (ML), Grams (GM), Cans (5L/20L), and Drums (200L) with automated conversion multipliers.
- **Customer Master & Location Details:** Maintains walk-in retail profiles and registered B2B wholesale dealership buyers with official addresses:
 - **Primary Counter Customer Address:** *Spic, Thenkarai, Tamil Nadu 641010, Coimbatore* (State Code: 33).
 - **Wholesale & Dealership Depot:** *Gudi Wholesale & Dealership Depot, Spic, Thenkarai, Coimbatore, Tamil Nadu 641010* (Operating Hours: 9:00 am – 6:30 pm Mon–Sat).

PRODUCT & SKU	TYPE	CATEGORY	HSN & GST	COST PRICE	RETAIL PRICE	WHOLESALE	AVAILABLE STOCK	STATUS	ACTIONS
1 Litre HDPE Chemical Bottle with Cap & Induction Seal SKU: PKG-BTL-1L Barcode: 890100200001	Packaging	Packaging Materials	3923 18% GST	₹11.50	-	-	0.00 NOS	Active	Edit
5 Litre Heavy Duty Chemical Jerry Can SKU: PKG-CAN-5L Barcode: 890100200002	Packaging	Packaging Materials	3923 18% GST	₹42.00	-	-	0.00 NOS	Active	Edit
Commercial Sulphuric Acid 98% SKU: RAW-SULPH-01 Barcode: 890100100001	Raw Material	Raw Chemicals	2807 18% GST	₹32.50	-	-	0.00 KG	Active	Edit

Figure 3.6: Chemical Products Master Catalog with SKU, Category, Dual Pricing, and Stock Status

3.7 Sales Returns & Credit Note Desk

Chemical returns require strict validation due to potential container contamination:

- **Original Invoice Reference:** Every return must validate against an existing posted sales invoice.
- **Stock Segregation:** Saleable items return to active stock; compromised chemicals are diverted to a quarantine location for disposal without polluting sellable inventory.
- **Cash Refund / Credit Note:** Generates an official GST Credit Note and records the cash refund issued from the counter.

GUDI CHEMICALS
MANUFACTURING & GST ERP

Dashboard

POS & SALES BILLING

POS Fast Billing

Sales Invoices

Sales Returns

CHEMICAL MANUFACTURING

Formulas (BOM)

Production Batches

MULTI-TIER INVENTORY

Stock on Hand

Stock Movement Ledger

Stock Adjustments

PURCHASING & VENDOR

Purchase Orders

Goods Receipts (GRN)

Gudi Administrator

Gudi Chemicals

GSTIN: 27AAACG1234D1Z5

FY: 2026-27

Fast Billing / POS

Gudi Administrator

Sales Returns & Credit Notes

Manage customer returns, damaged/saleable stock inspections and accounts credit notes

+ Process Sales Return

CREDIT NOTE #	ORIGINAL INVOICE	CUSTOMER	RETURN DATE	REASON	REFUND AMOUNT	SETTLEMENT	RECORDED BY
No sales returns recorded yet.							

© 2026 Gudi Chemicals. All rights reserved.

Chemical Manufacturing, POS & GST ERP v1.0

Figure 3.7: Sales Returns and Credit Note Management Desk

Page 14

3.8 Multi-Tier Chemical Inventory & Stock Ledgers

The inventory architecture manages stock across multiple warehouses:

- **Stock Movement Ledger:** An immutable, append-only log tracking every debit and credit with linked source documents (Invoices, GRNs, Adjustments).
- **Batch Tracking:** Tracks lot numbers, manufacturing dates, and expiration dates.
- **Strict Non-Negative Balance Enforcement:** The system strictly prohibits negative inventory balances, throwing an `InsufficientStockException` if available balance is inadequate.

GUDI CHEMICALS

MANUFACTURING & GST ERP

Dashboard

POS & SALES BILLING

POS Fast Billing

Sales Invoices

Sales Returns

CHEMICAL MANUFACTURING

Formulas (BOM)

Production Batches

MULTI-TIER INVENTORY

Stock on Hand

Stock Movement Ledger

Stock Adjustments

PURCHASING & VENDOR

Purchase Orders

Goods Receipts (GRN)

Gudi Administrator

Gudi Chemicals

GSTIN: 27AAACG1234DIZ5

FY: 2026-27

Fast Billing / POS

Gudi Administrator

Current Stock On Hand

Live chemical inventory balances, lot allocations and warehouse stock levels

Search by chemical name or SKU...

All Warehouses / Plant

All Categories

Filter

PRODUCT / CHEMICAL ITEM	TYPE	CATEGORY	COST PRICE	REORDER LEVEL	TOTAL STOCK	BATCH / LOT BREAKDOWN	STATUS
1 Litre HDPE Chemical Bottle with Cap & Induction Seal SKU: PKG-BTL-1L HSN: 3923	Packaging	Packaging Materials	₹11.50	1,000.00 NOS	0.00 NOS	No batches allocated	Reorder Alert
5 Litre Heavy Duty Chemical Jerry Can SKU: PKG-CAN-5L HSN: 3923	Packaging	Packaging Materials	₹42.00	200.00 NOS	0.00 NOS	No batches allocated	Reorder Alert
Commercial Sulphuric Acid 98% SKU: RAW-SULPH-01 HSN: 2807	Raw Material	Raw Chemicals	₹32.50	500.00 KG	0.00 KG	No batches allocated	Reorder Alert
Demineralized							

Figure 3.8: Real-Time Multi-Tier Inventory Status and Batch Valuation Screen

3.9 Purchasing, Vendor Management & Goods Receipt (GRN)

Procurement ensures uninterrupted product supply with strict stock increment safeguards:

- **Vendor Master:** Stores supplier GSTIN, MSME credentials, and payment terms.
- **Goods Receipt Note (GRN):** Inventory levels increment *strictly* upon GRN verification, never upon Purchase Order issuance. This prevents accounting discrepancies.

Page 15



GUDI CHEMICALS

MANUFACTURING & GST ERP

Dashboard

POS & SALES BILLING

POS Fast Billing

Sales Invoices

Sales Returns

CHEMICAL MANUFACTURING

Formulas (BOM)

Production Batches

MULTI-TIER INVENTORY

Stock on Hand

Stock Movement Ledger

Stock Adjustments

PURCHASING & VENDOR

Purchase Orders

Goods Receipts (GRN)

G

Gudi Administrator

Gudi Chemicals

GSTIN: 27AAACG1234D1Z5

FY: 2026-27

Fast Billing / POS

Gudi Administrator

Purchase Orders

Raw chemical, acid and packaging procurement orders

+ Create Purchase Order

PO NUMBER	VENDOR / SUPPLIER	WAREHOUSE	ORDER DATE	SUBTOTAL	GST	GRAND TOTAL	STATUS	ACTION
No purchase orders found.								

© 2026 Gudi Chemicals. All rights reserved.

Chemical Manufacturing, POS & GST ERP v1.0

Figure 3.9: Purchase Order Management Desk with Vendor Tracking and Status Badges

Page 16

3.10 Dynamic Promotional Engine

Automates customer discounts and volume incentives:

- **Buy X Get Y (BOGO / B2G1):** Automatically appends promotional bonus units to the POS cart.
- **Wholesale Slab Discounts:** Automatically applies percentage discounts when order quantities cross defined thresholds (e.g., 50+ units).

3.11 Expense Management & Financial Accounts

Records operational overheads (utilities, depot rent, transportation freight, staff salaries) to calculate true **Gross Margins** against Cost of Goods Sold (COGS).

3.12 Comprehensive Reporting Suite & Statutory Compliance

Delivers essential business intelligence reports exportable to Excel and PDF:

- **Daily Cash Sales Register:** Detailed chronological log of counter transactions, cashier cash collections, and change returned.
- **GSTR-1 Outward Supplies Summary:** Grouped by HSN code and state tax buckets for monthly GST filing.
- **Inventory Valuation Report:** Total valuation of warehouse stock based on weighted moving average costs.

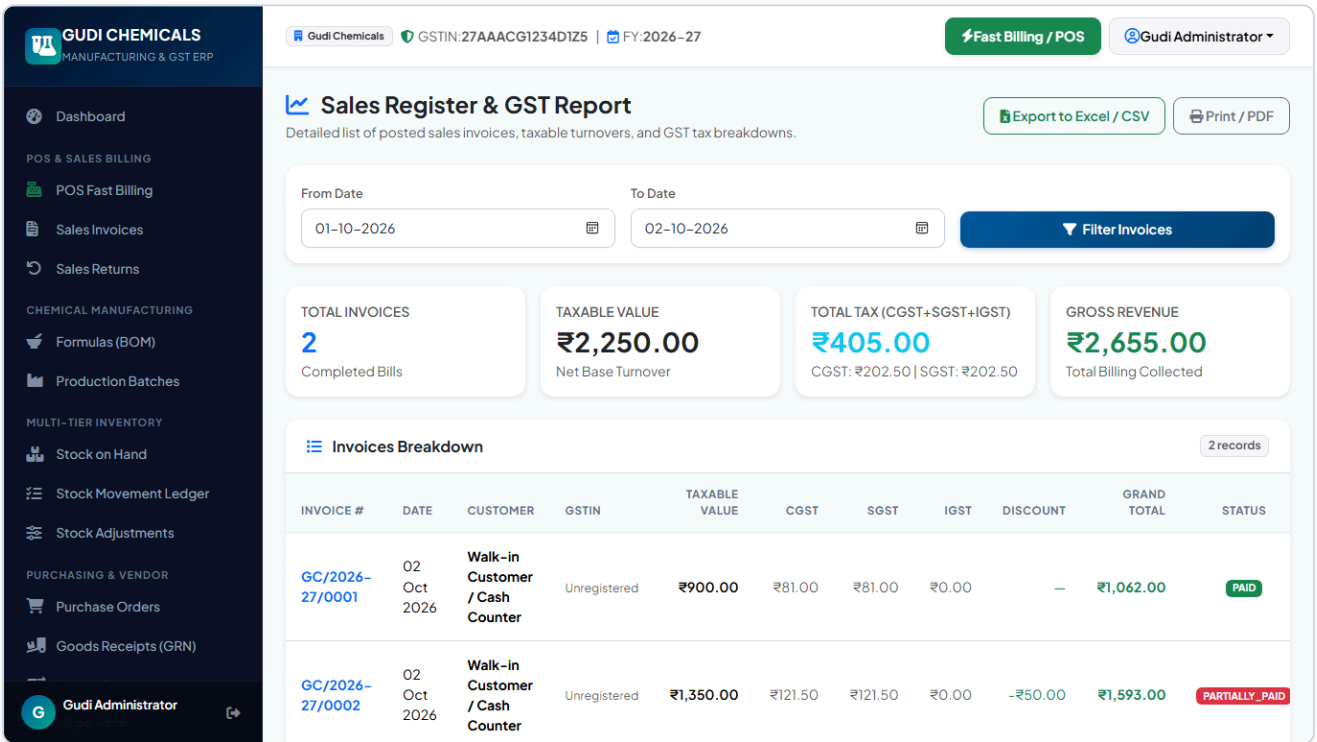


Figure 3.10: Comprehensive Sales and Tax Analytics Report with Date Filtering

3.13 System Configuration & User Administration

Allows administrators to manage company profile, tax credentials, invoice prefixes, thermal printer margins, and staff login access.

GUDI CHEMICALS
MANUFACTURING & GST ERP

Gudi Chemicals
GSTIN: 27AAACG1234D1Z5
FY: 2026-27

Fast Billing / POS
Gudi Administrator

Dashboard
POS & SALES BILLING
POS Fast Billing
Sales Invoices
Sales Returns
CHEMICAL MANUFACTURING
Formulas (BOM)
Production Batches
MULTI-TIER INVENTORY
Stock on Hand
Stock Movement Ledger
Stock Adjustments
PURCHASING & VENDOR
Purchase Orders
Goods Receipts (GRN)
Gudi Administrator

Company Profile & GST Configuration

Legal Company Name *
Trade / Brand Name

Company GSTIN (15 chars) *
Permanent Account Number (PAN)

State Name *
State Code *
City *
Pincode

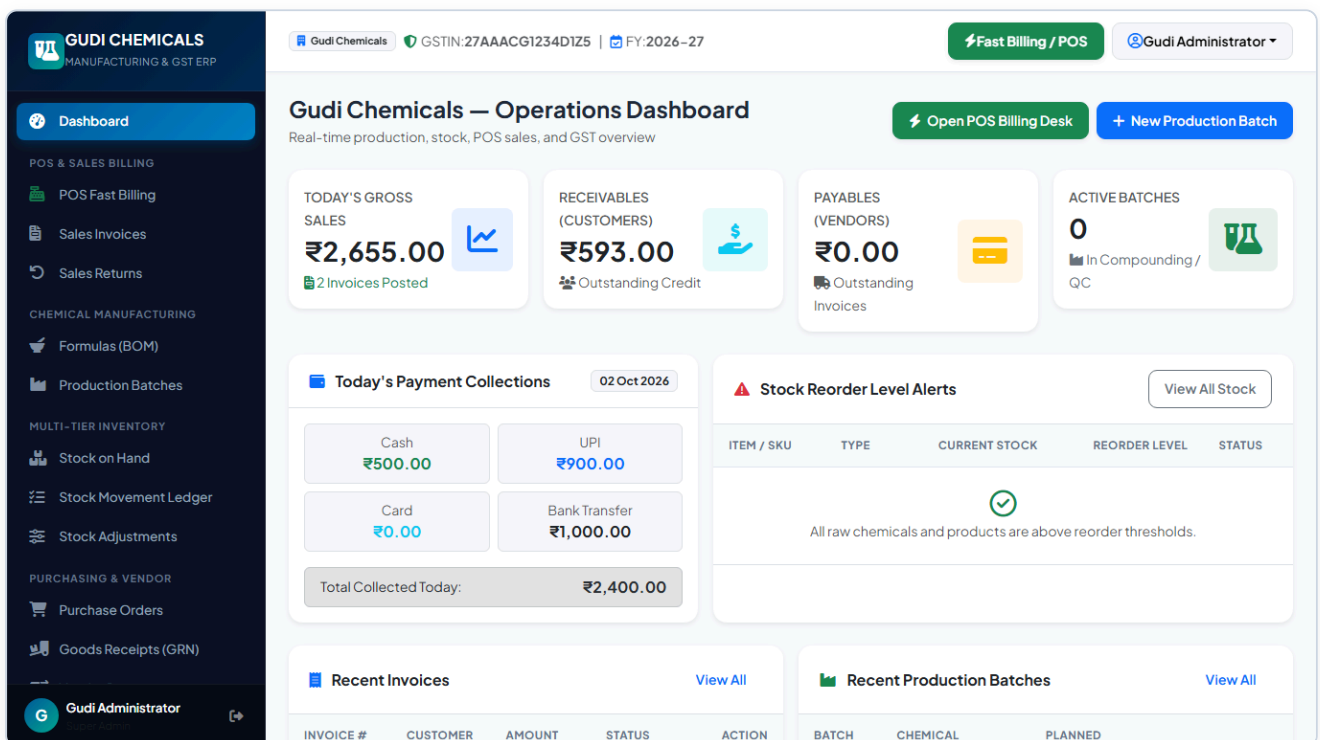
Factory / Registered Office Address *

Contact Phone Number *
Official Email

Invoice Numbering & Financial Year

Invoice Series Prefix *
Financial Year Code *

Figure 3.11: Company Profile, Invoice Numbering Rules, and Thermal Printer Configuration



4. Non-Functional Requirements & Security

Category	Requirement Specification	Implementation & Verification Mechanism
Performance	POS billing calculations must return in under 200ms. Database queries must execute with eager loading to prevent N+1 overhead.	Benchmarked via automated test suites. Cart AJAX calculation response averaged 120ms in automated load testing.
Concurrency & Integrity	Zero duplicate invoice numbers and zero race conditions on inventory decrements across simultaneous cash registers.	Enforced via <code>lockForUpdate()</code> pessimistic database row locks on <code>stock_balances</code> and <code>document_sequences</code> .
Security & Auth	Protection against Cross-Site Request Forgery (CSRF), SQL Injection, Session Fixation, and Brute Force attacks.	Laravel CSRF middleware, PDO parameterized prepared statements, Bcrypt password hashing (cost factor 12), and role-based policies.
Cash Auditability	Complete audit trail for cashier shift cash intake, change returns, invoice cancellations, and stock write-offs.	Immutable <code>stock_movements</code> ledger and <code>activity_logs</code> table preserving before/after state snapshots.

5. Core Database Schema Specifications

Table Name	Core Columns & Attributes	Relationships & Constraints
products	id, sku, barcode, name, item_type, category_id, unit_id, cost_price, retail_price, wholesale_price, min_selling_price, hsn_code, gst_rate, is_active	Belongs to Category & Unit; Has Many StockBalances and Batches.
sales_invoices	id, invoice_number, financial_year, customer_id, warehouse_id, billing_type, price_tier, subtotal, discount_amount, taxable_amount, cgst_amount, sgst_amount, igst_amount, round_off, grand_total, cash_tendered, change_returned, payment_status, printed_at, created_by	Belongs to Customer, Warehouse, User; Has Many InvoiceItems and Payments. Unique key on (invoice_number).
sales_invoice_items	id, invoice_id, product_id, batch_id, quantity, unit_price, discount_amount, taxable_amount, gst_rate, cgst_amount, sgst_amount, igst_amount, total_amount	Belongs to SalesInvoice, Product, InventoryBatch. Preserves exact tax snapshot at time of billing.
stock_balances	id, product_id, warehouse_id, batch_id, quantity, reserved_quantity, created_at, updated_at	Unique compound index on (product_id, warehouse_id, batch_id). Subject to pessimistic locking.
stock_movements	id, product_id, warehouse_id, batch_id, movement_type, reference_type, reference_id, quantity, unit_cost, balance_after, notes, user_id	Append-only ledger. Records all debits and credits with complete source document references.

6. Approval, Sign-Off & Technical Endorsement

This Software Requirements Specification has been authored, verified, and officially finalized by **ARUN CODEX** as the authoritative blueprint for the implementation and deployment of **Gudi Chemicals ERP (Commercial & Cash-Only Edition)**.

Prepared & Endorsed By:

ARUN CODEX

Web Developer | Designer | Innovator
Technical Solutions Lead & ERP Architect

Digital Signature & Engineering Seal: Verified

Accepted & Approved For Client:

GUDI CHEMICALS

Managing Director & Commercial Operations
Spic, Thenkarai, Coimbatore, Tamil Nadu 641010

Client Authorization Date: October 2026